CIS 635 Knowledge Discovery & Data Mining

Introduction to Unsupervised Learning

Supervised Learning

- We learned about **Classification** and **Regression**
- These are examples of **supervised** learning

Supervised (Discriminative) Learning

$$f(y='diabetes' | X=['age', 'bmi'])$$

Supervised (Discriminative) Learning

$f(y='diabetes'|X=['age', 'bmi'])$

$f(y|X)$

Essentially we are learning the relationship between X and y

Supervised (Discriminative) Learning

$y \in R$

- *Insurance cost : $y \in R$*
- *House price: $y \in R$*
- *Weather prediction: $\boldsymbol{Y \in R}$*
- *Energy consumption: $y \in R$*
- *Sales forecasting: $y \in R$*

Regression

$y \in \{....\}$

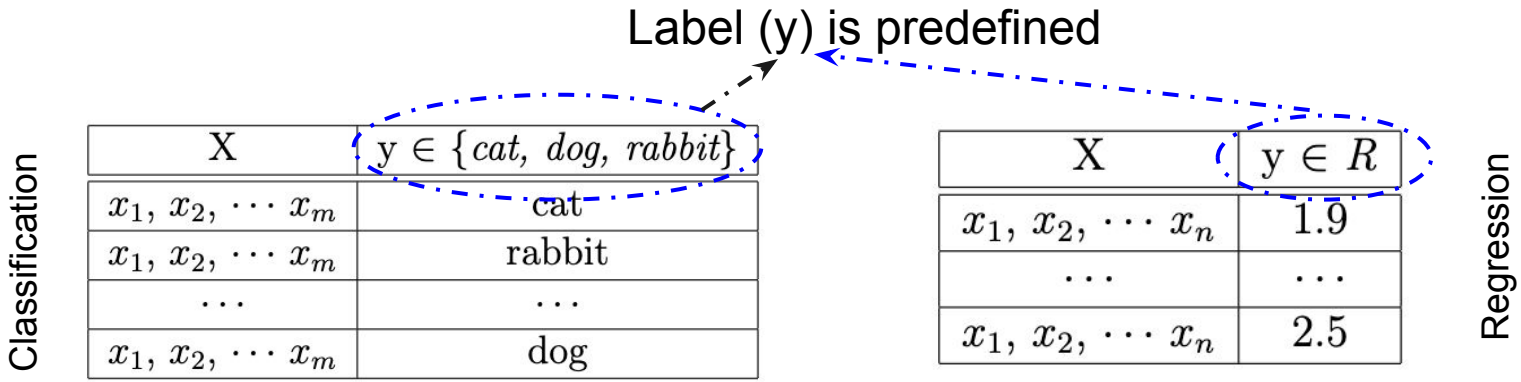
- *Cat vs dog: $y \in \{cat, dog\}$*
- *Spam filter: $y \in \{spam, not\text{-}spam\}$*
- *Digit classification: $y \in \{0,1,....9\}$*
- *Sentiment classification: $y \in$ {happy, sad, confused, angry …}*

note: has to be from a closed set

Classification

Supervised Learning

- We learned about **Classification** and **Regression**
- These are examples of **supervised** learning
- In your data you have both **X(features)** and **y(Labels)**



Unsupervised Learning

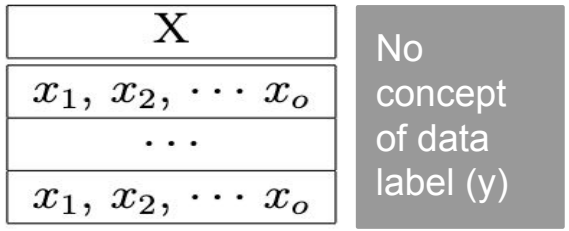
- In contrast, in **Unsupervised learning**, we have to learn meaningful **representations/models** from **X(features)** only.
- **Clustering** is an example of unsupervised learning

X
$x_1, x_2, \cdots x_o$
$\cdots$
$x_1, x_2, \cdots x_o$

No concept of data label (y)

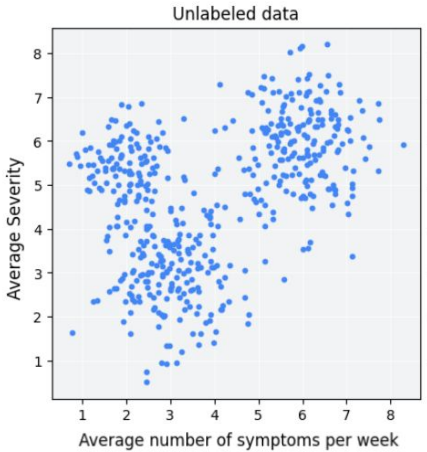
Unsupervised Learning

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- **Clustering** is an example of unsupervised learning



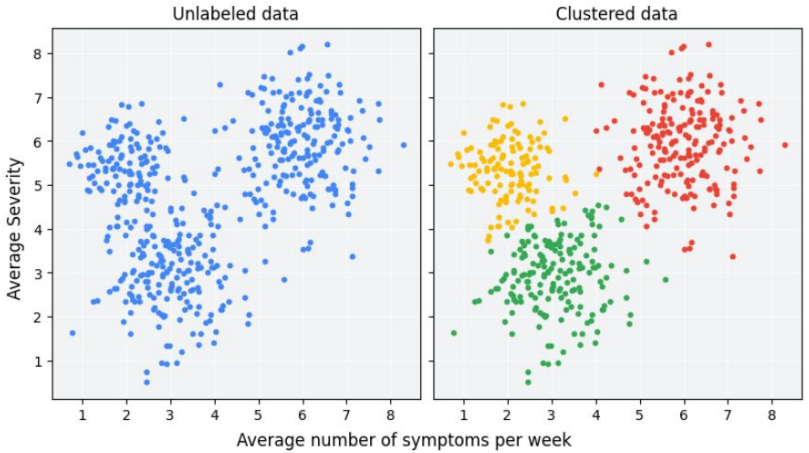
Clustering

- A medical study involving
  - Average number of symptoms per week
  - Average severity



Clustering

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Clustering

- What is clustering?
- **Clustering algorithms:**
  - **K-Means**: Centroid Based
  - **Hierarchical clustering**: Distance connectivity based
  - **GMM**: Distribution based
  - **DBSCAN**: Density Based

- Identifying the number of clusters ?

What is clustering ?

- Grouping of data points based on some features (**X**). Group **indices** are the output

- An example of unsupervised learning

We can use features such as **X {size, color}** to group apples on the right



https://www.codeproject.com/Articles/1273113/Apple-tron-an-AI-for-farmers

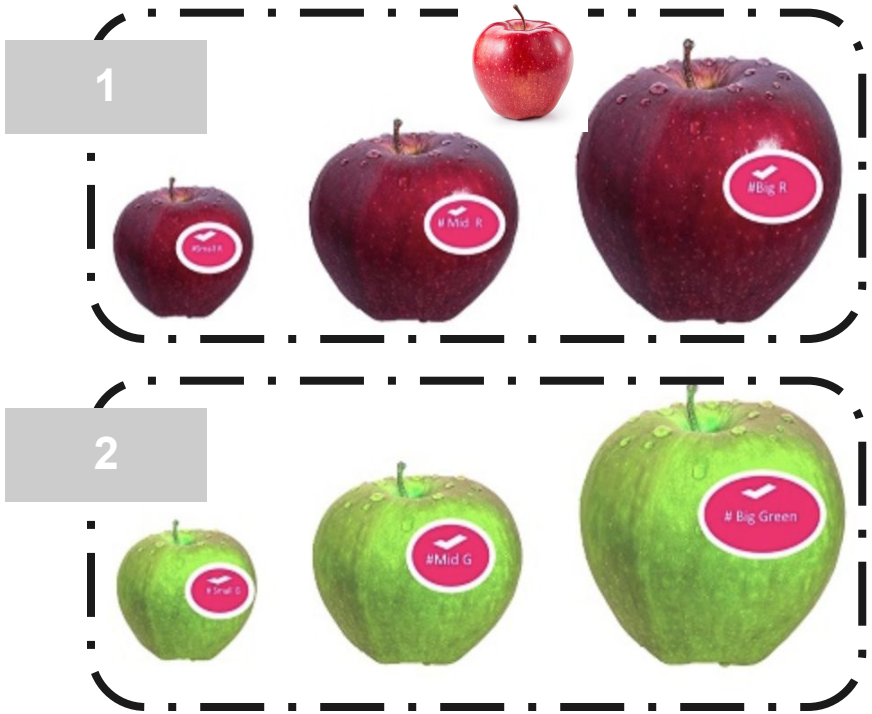
What is clustering ?

- Grouping of data points based on some features (**X**). Group **indices** are the output

- An example of unsupervised learning

- **Labeling a test case:**

  Output: group index **1**



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Based on color

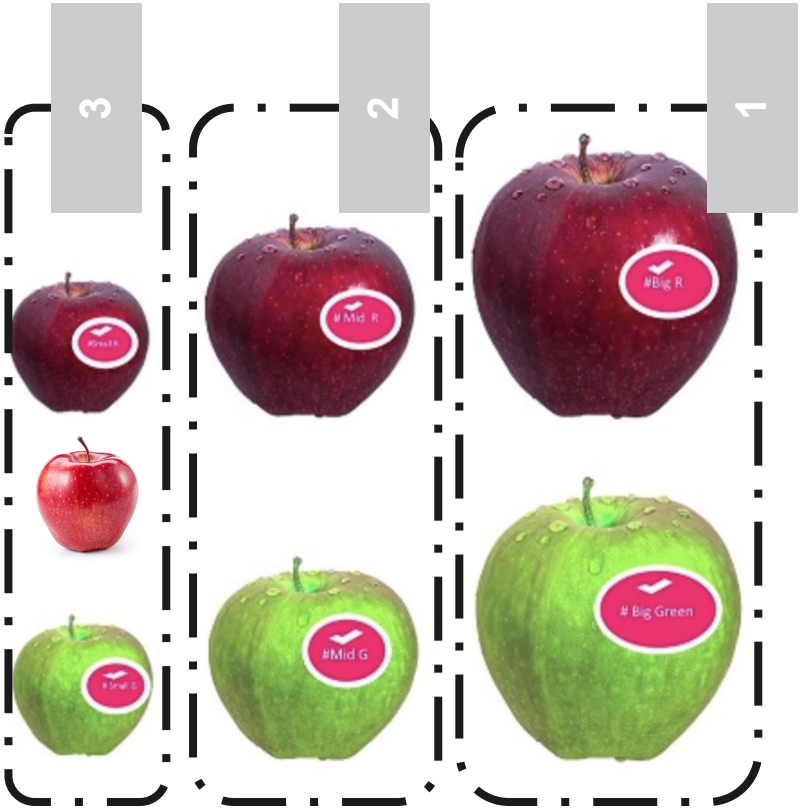
What is clustering ?

- Grouping of data points based on some features (**X**). Group **indices** are the output

- An example of unsupervised learning

- **Labeling a test case:**

  Output: group index **3**



https://www.codeproject.com/Articles/1273113/Apple-tron-an-AI-for-farmers

Based on size

What is clustering ?

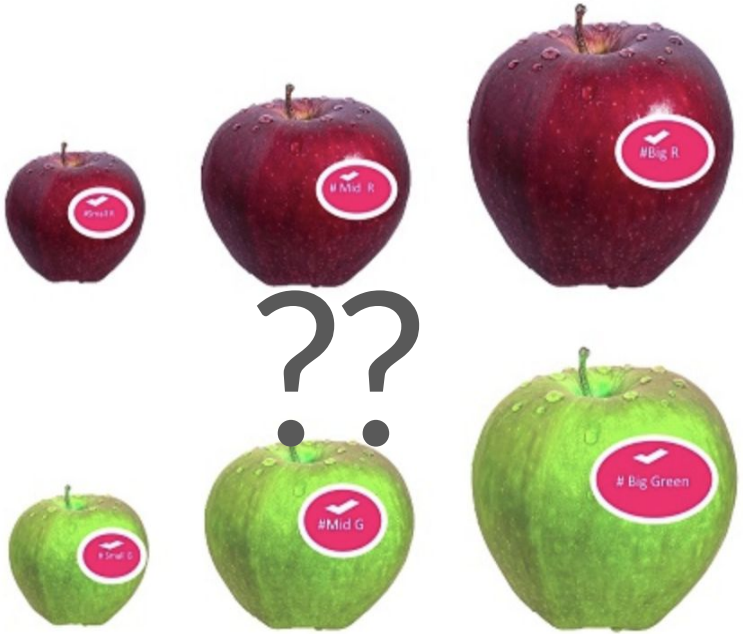
What if we want use both features: size and color?

What is clustering ?

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- An example of unsupervised learning

We can use features such as **X {size, color}** to group apples on the right



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Usages of Clustering

Clustering has a myriad of uses in a variety of industries. Some common applications for clustering include the following:

- market segmentation
- social network analysis
- search result grouping
- medical imaging
- image segmentation
- anomaly detection

ref

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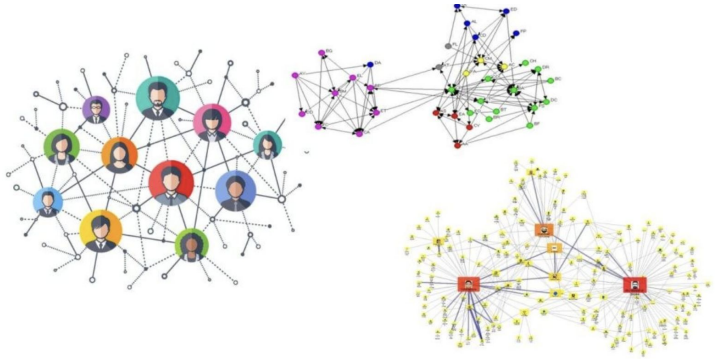


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